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Exploring Jet Substructure in Public LHC Data

Alexandre Maria Neves de Barbosa Mendonça

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Orientador(es)

Professor Doutor José Guilherme Teixeira de Almeida Milhano
Professor Doutor Nuno Filipe da Silva Fernandes de Castro

Júri

Presidente: Professor Doutor Nome Completo

Orientador: Professor Doutor José Guilherme Teixeira de Almeida Milhano
Professor Doutor Nuno Filipe da Silva Fernandes de Castro

Vogais: Professor Doutor Nome Completo
Doutor Nome Completo

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AGRADECIMENTOS

RESUMO

Palavras-chave:

ABSTRACT

Keywords:

LIST OF ABBREVIATIONS AND SYMBOLS

1 INTRODUCTION

1.1 Context and motivation

The Large Hadron Collider (LHC) at CERN delivers the highest energy proton-proton collisions ever produced in a laboratory and, with them, an enormous sample of strongly interacting final states. The majority of these collisions are governed by Quantum Chromodynamics (QCD), the gauge theory of the strong interaction. The study and comprehension of these collisions are not only a prerequisite for suppressing backgrounds in searches for new phenomena but also the structure of the hadronic final state can be examined and compared to theory directly, at scales where the coupling is weak enough for perturbation theory to be applied and yet the dynamics still remain rich.

The most prominent objects in these final states are jets. Jets are collimated sprays of hadrons that emerge from the fragmentation of high energy quarks and gluons. A jet is not a fundamental object but instead a definition. A clustering algorithm specifies which detector inputs are grouped together and a recombination scheme specifies how momentum is assigned to a result. This defines a jet, from which an interesting and important question then arises. What does the radiation inside the jets look like? This domain is called jet substructure, which over the past decade has grown from a collection of tagging tools into a precision probe of QCD dynamics in its own right.

In order to organise and look at the internal structure of a jet, one can use the Lund jet plane [1]. The Lund jet plane is a two dimensional map of a jet's branching history on which soft, collinear and non perturbative regions of QCD radiation occupy distinct territories. This map records where the jet's internal emissions occur, but it sets aside an intriguing piece of information, the orientation of each splitting relative to the one before. This orientation is not random. Through the linear polarisation of intermediate gluons, successive splittings are correlated in their relative azimuthal angle, producing a characteristic modulation in the distribution of that angle. This is a genuine quantum interference effect which survives only because the intermediate gluon is described by an amplitude rather than a probability, and it is absent from any picture that treats successive emissions as independent classical branchings.

These collinear spin correlations are well understood theoretically and are implemented in modern parton shower algorithms, yet establishing them cleanly in collision data has proved rather difficult. The reason is that the effect carries opposite signs in different splitting channels. The $g \to q\bar{q}$ and $g \to gg$ channels contribute with opposite sign, so a measurement that sums all channels sees the two contributions partially cancel. This interference is not absent from such an inclusive measurement. It is diluted and the apparent size reflects the accidental flavour composition of the sample rather than the strength of the underlying physics.

This is the obstacle the present work addresses. A meaningful measurement of the effect requires knowing which splitting produced each emision. However, the flavour of an individual splitting inside a jet is not directly observable and must be inferred. Working entirely on public CERN Open Data [2], more precisely with CMS Open Data [3], this dissertation builds a reproducible pipeline from the released samples through Lund declustering to the azimuthal spin observable, and introduces a flavour selection that isolates the $g \to q\bar{q}$ channel. The resulting modulation is reported not as a single number but as a function of channel purity achieved by that selection, a presentation that separates the physics of the

interference from the performance of the selection and allows the magnitude of the effect to be read off in the limit of a pure sample.

The Lund declustering angle $\Delta\psi_{12}$ is not the only way to reach this effect. The same $\cos 2\psi$ modulation also imprints itself on the flow of energy within a jet, and can be accessed through a three-point energy correlator, which probes the relative azimuthal orientation of energy deposited by triplets of particles without reference to any clustering history. The two observables are complementary: one reads the effect from the reconstructed branching sequence, the other from the energy flow directly, and each carries a different exposure to the choices and imperfections of the measurement. This work develops both, treating the Lund observable as the primary measurement and the energy correlator as an independent cross-check on the same underlying physics.

The fact that this analysis rests on public data is not incidental. The CERN Open Data Portal [2] has made it possible for work outside the experimental collaborations to produce meaningful results and to carry out genuinely new measurements. The present work is intended as a contribution in that spirit. It builds an end to end pipeline from a single CMS dataset to the final measurements, establishing along the way a workflow for accessing, handling and measuring with CMS Open Data.

1.2 Objectives

The main aim of this dissertation is to measure collinear spin correlations in QCD jets using public LHC data and to establish the flavour selection that such a measurement requires in order to be meaningful. For the purpose of achieving that goal, the work is organised around a sequence of objectives. The first is to build a reproducible pipeline over CERN Open Data samples capable of extracting the necessary variables to begin a jet substructure analysis. This means that the data samples must contain jet constituent information, thus the data format chosen must retain that level of detail, ruling out the more compact samples in favour of one that preserves per-constituent quantities.

With that pipeline in place, the second objective is to construct the primary and secondary Lund jet planes and to verify that they reproduce the known features of QCD radiation, since this agreement is what establishes that the declustering is sound before any finer observable is built on top of it. Following that, the third objective is to implement the azimuthal observable $\Delta\psi_{12}$ between the hardest primary and secondary branchings, and to study its distribution both with and without grooming, recovering the inclusive result before attempting to sharpen it.

The fourth objective is the central one, and the point at which this work departs from a purely inclusive analysis. Because the spin-interference signal carries opposite signs in the $g \to q\bar{q}$ and $g \to gg$ channels, an inclusive measurement dilutes it. Isolating a single channel is therefore not a refinement but a precondition for measuring the effect at its true strength. This objective is thus to develop a flavour selection capable of enriching the $g \to q\bar{q}$ channel, and to quantify both how well it performs and how pure a sample it can reach in principle. The fifth objective builds directly on it. It is to measure $\langle \cos 2\psi_{12} \rangle$ as a function of the channel purity delivered by that selection and to establish the value characteristic of a pure $g \to q\bar{q}$ sample.

Finally, the work aims to assess the systematic uncertainties affecting this measurement and to identify the dominant limitations of the approach, so that the extensions it opens up can be set out on solid ground.

1.3 Thesis outline

Chapter 2 develops the theoretical material on which the measurement rests, such as the elements of QCD relevant to jet formation, the definition and reconstruction of jets and the algorithms used to build and decluster them, the Lund jet plane, and the origin of collinear spin correlations together with the two observables through which this work accesses them. Chapter 3 turns to the data and the analysis, and covers the CERN Open Data resources and the CMS detector, the data format and dataset used, and the pipeline that runs from the released samples through Lund declustering to the azimuthal observable, the flavour selection, and the treatment of systematic uncertainties. Chapter 4 presents the validation of the pipeline and the measurement itself, and discusses the results in the light of the expectations set out in Chapter 2. Finally, Chapter 5 returns to the objectives of Section 1.2, states what has been established and with what confidence, and outlines the extensions that the present framework makes possible.

2 THEORETICAL BACKGROUND

2.1 Quantum Chromodynamics

Quantim Chromodynamics (QCD) is the quantum field theory of the strong interaction, a non-Abelian gauge theory built on the colour symmetry group $SU(3)$ [4]. Its fundamental degrees of freedom are quarks, which carry colour charge, and gluons, the gauge bosons that mediate the interaction. The non-Abelian character of the theory has a decisive consequence absent from Quantum Electrodynamics. The gluons themselves carry colour, and therefore couple directly to one another. To a large extent, the dynamics of the strong interaction are the dynamics of this gluon self-coupling, and the two properties of QCD that govern everything in this chapter both follow from it.

2.1.1 Asymptotic freedom and confinement

The first property is asymptotic freedom. The strong coupling α_s decreases as the momentum transfer of an interaction increases, so that at the high scales probed in LHC collisions the quarks and gluons interact weakly and perturbation theory applies [5, 6]. The running of the coupling is captured at leading order by

$$\alpha_s(Q^2) = \frac{\alpha_s(\mu^2)}{1 + \alpha_s(\mu^2)\,\frac{33 - 2n_f}{12\pi}\,\ln\frac{Q^2}{\mu^2}}, \tag{2.1}$$

where Q is the scale of the process, μ a reference scale, and n_f the number of active quark flavours. The positive sign of the coefficient $33 - 2n_f$, which holds for any physical number of flavours, is what makes the coupling fall at large Q. It is a direct imprint of the gluon self-interaction, and its discovery established that a peturbative treatment of hard processes is justified. It is convenient to trade the reference scale for a single dimensionful constant Λ_{QCD}, the scale at which the peturbative coupling formally diverges, of an order of a few hundered MeV. In terms of it the coupling is small for $Q \gg \Lambda_{\text{QCD}}$ and grows strong as Q approaches it.

This growth is the second property called confinement. At low momentum transfer α_s becomes large and colour-charged particles cannot be isolated. A quark or gluon produced in a hard scattering does not propagate freely to a detector, because as it separates from other colour charges, the energy stored in the colour field between them grows with distance until it becomes energetically favourable to create a new quark-antiquark pairs from the vacuum. The original parton is thereby dressed into a spray of colour-neutral hadrons. This is why the strongly interacting final state reaches the detector not as partons but as the collimated sprays introduced in Chapter 1, and why any comparison between a peturbative calculation and a measurement must bridge the divide between the patronic and hadronic descriptions.

2.1.2 Soft and collinear radiation

The bridge between the hard parton and the observed hadrons is built from successive emissions, and the pattern of those emissions is far from uniform. When a parton radiates a gluon, the emission probability

at first order in the coupling takes the schematic form

$$d\mathcal{P} \propto \frac{2\alpha_s C}{\pi} \frac{d\theta}{\theta} \frac{dE}{E}, \tag{2.2}$$

where E and θ are the energy and emission angle of the radiated gluon and C is a colour factor. The two factors carry the characteristic structure of QCD radiation. The factor $d\theta/\theta$ diverges as $\theta \to 0$ meaning that emmissions are enhanced when the gluon is radiated close to the direction of its emmiter. This is known as the collinear divergence. The factor dE/E diverges as $E \to 0$ meaning that emissions are enhanced when the gluon is soft. This is known as the soft divergence. Together they mean that a hard parton radiates predominantly softly and at small angles, and that number of such emissions is, in a precise logarithmic sense, large.

The collinear structure is described more completely by the Dokshitzer–Gribov–Lipatov–Altarelli–Parisi (DGLAP) splitting functions [4], which give the probability that a parton of one type splits into two, as a function of the momentum fraction z carried by one of the products. For the three splittings relevant to this work they read, at leading order,

$$P_{q\to qg}(z) = C_F \frac{1+z^2}{1-z}, \tag{2.3}$$

$$P_{g\to gg}(z) = C_A \left[\frac{z}{1-z} + \frac{1-z}{z} + z(1-z)\right], \tag{2.4}$$

$$P_{g\to q\bar{q}}(z) = T_R \left[z^2 + (1-z)^2\right], \tag{2.5}$$

with colour factors $C_F = 4/3$, $C_A = 3$ and $T_R = 1/2$. The distinction between the gluon splitting channels of Eqs. (2.4) and (2.5), one producing a pair of gluons and the other a quark-antiquark pair, will prove central. The two channels differ not only in these rates but in the spin structure of their emissions, and it is that difference which the spin observables of Section 2.4 are designed to probe.

These divergences are not a pathology. They are integrable, and in any physical, infrared-safe observable the divergent real emissions cancel against corresponding virtual corrections, leaving a finite result. This cancellation is the requirement that shapes the very definition of a jet in Section 2.2. Their logarithmic structure, however, is physical and measurable, and it is exactly the structure that the Lund jet plane of Section 2.3 is built to expose. By mapping each emision onto logarithmic axes of angle and transverse momentum, it turns the double logarithmic enhancement of equation (2.2) into an approximately uniform density, against which finer correlations stand out.

2.1.3 Parton showers and hadronization

Because a single hard parton typically radiates many times before hadronizing, the emissions can be organised into a sequence orderred from the hardest, widest-angle splittings down to the soft and collinear ones. This sequence is what is known as a parton shower. Modern Monte Carlo event generators implement this sequence probabilistically through the splitting functions of equations (2.3)-(2.5), evolving each parton down from the hard scale of the collision to a low scale of order 1 GeV, at which the perturbative description breaks down and confinement takes over. Below that scale the partons are converted into hadrons by phenomenological hadronization models, tuned to data rather than derived from first principles.

This picture, a hard, perturbatively calculable core, dressed by a shower of increasingly soft and collinear emissions and finally hadronized, underlies the whole of the analysis that follows. The observables studied in this work are built to read the perturbative shower through the hadrons that reach the detector, and for this reason the measurement is deliberately confined to a hard kinematic regime, where the perturbative component dominates and non-perturbative corrections are comparatively suppressed.

2.2 Jets

As established in Section 2.1, a coloured parton produced in a hard scattering does not reach the detector directly. It radiates and hadronizes, arriving as a collimated spray of hadrons. A jet is the operational device that inverts this process approximately, grouping those hadrons back together so that their combined momentum estimates that of the parton that produced them. A jet is therefore not a physical object with a sharp definition but a construction, and its properties depend on the prescription used to build it. This section sets out that prescription: the requirement any jet definition must satisfy to be usable in perturbative QCD, and the three algorithms, built on that requirement, on which this work relies.

2.2.1 Jet algorithms and infrared–collinear safety

Two ingredients specify how a jet is built. A jet algorithm decides which inputs, from detector-level energy deposits, reconstructed particles, or, in a calculation, partons, are grouped together into a jet. A recombination scheme specifies how the momenta of those grouped inputs are combined into the momentum of the jet. Throughout this work the inputs are the particle-flow candidates described in Chapter 3, and momenta are combined in the E-scheme, in which four-momenta are simply added. This scheme is Lorentz invariant and yields jets with non-zero mass.

Unfortunately, not every conceivable grouping prescription is admissible. The requirement that disqualifies most of them is infrared and collinear (IRC) safety. An observable is infrared safe if its value is unchanged by the addition of an arbitrarily soft emission, and collinear safe if it is unchanged when one of its inputs is replaced by two collinear inputs sharing its momentum. A jet algorithm is IRC safe when the set of jets it returns is stable under both these operations. A soft particle must not create or destroy a jet and a collinear splitting must not change how particles are assigned to jets.

The reason why this requirement is not optional follows directly from Section 2.1. The emission probability of equation (2.2) diverges in both the soft and the collinear limits. In any IRC-safe observable these divergent real emissions cancel against equally divergent virtual corrections, leaving a finite result. In an IRC-unsafe observable, that cancellation is spoiled and a fixed-order peturbative prediction returns infinity rather than a number. An IRC-unsafe jet definition is therefore not merely inconvenient but it also cannot be compared with peturbative QCD at all. Since the purpose of measuring jets is to confront the theory with data, IRC safety is a precondition and not a refinement.

This criterion has been decisive in the historical development of jet algorithms. The earliest definitions were cone algorithms, which grouped particles falling within a fixed angular radius of a candidate axis. In their simplest, seed-based forms these algorithms were not IRC safe because a soft particle could seed a new candidate cone, or a collinear splitting could change which particles acted as seeds, and either could alter the resulting jets. Much effort was devoted to repairing them, and eventually producing seedless variants that restored collinear safety. The alternative family, the sequential recombination algorithms, are

IRC safe by construction, and it is to this family that algorithm used in this work belongs.

It is worth making concrete how such a failure arises, since the same mechanism will reappear when flavour is assigned to jets in Section 2.2.4. Consider a seed-based cone algorithm applied to an event with two hard particles whose separation is slightly greater than the cone radius, so that each seeds its own jet. The addition of a single soft particle between them may seed a third cone that overlaps both, merging the two hard particles into one jet where before there were two which, again, it is an infrared-unsafe change, since an arbitrarily soft emission has altered the hard jets. A collinear splitting produces the collinear-unsafe failure. Replacing one hard seed by two collinear particles, each carrying half its transverse momentum, may leave neither above the seed threshold, so that a jet which existed before the splitting is absent after it. In both cases, the divergent emissions of equation (2.2) feed directly into the jets themselves, and the perturbative prediction diverges with them.

The sequential recombination algorithms introduced below evade both failures by never relying on seeds or on fixed geometric cones. They build jets by repeatedly merging inputs according to a distance measure, so that soft or collinear addition changes only the order in which already nearby inputs are combined, never the final assigment of hard momentum to jets. This structural difference is the reason the entire famiky is IRC safe, and it is what recommends it over the cone algorithms for any measurement to be compared with theory.

A sequential recombination algorithm is defined by a distance measure between each pair of inputs and between each input and the beam. The generalised-k_T family [7] writes these as

$$d_{ij} = \min\left(p_{\mathrm{T},i}^{2p},\, p_{\mathrm{T},j}^{2p}\right) \frac{\Delta R_{ij}^2}{R^2}, \tag{2.6}$$

$$d_{iB} = p_{\mathrm{T},i}^{2p}, \tag{2.7}$$

where $\Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$ is the squared angular separation of the inputs in rapidity y and azimuth ϕ, R is the jet radius parameter, and the integer p selects the algorithm. At each step the algorithm finds the smallest of all the distances. If it is a d_{ij}, the inputs i and j are recombined into a single input. If it is a d_{iB}, the input i is declared a final jet and removed. The procedure repeats until no inputs remain. The set of jets returned is IRC safe for any integer value of p, but the value of p changes profoundly how the clustering proceeds. The three choices $p = -1$, $p = 0$ and $p = +1$ give, respectively, the anti-k_T, Cambridge/Aachen and k_T algorithms, of which the first two are the tools this work relies on and are described in the following subsections.

Common to all three is the radius parameter R, which sets the angular reach of the clustering through the ratio $\Delta R_{ij}^2/R^2$ in equation (2.6). A smaller R yields narrower jets that resolve nearby structure but capture less of a single parton's radiation. A larger R collects more of that radiation at the price of admitting more contamination from unrelated soft activity. The appropriate choice depends on the use to which the jets are put, and the two algorithms this work relies on are used with different values of R for correspondingly different purposes. This point is, again, taken up in Sections 2.2.2 and 2.2.3.

2.2.2 The anti-k_T algorithm

The choice $p = -1$ in Eqs. (2.6) and (2.7) defines the anti-k_T algorithm [8]. With this value the inter-particle distance becomes

$$d_{ij} = \min\left(\frac{1}{p_{\mathsf{T},i}^2}, \frac{1}{p_{\mathsf{T},j}^2}\right)\frac{\Delta R_{ij}^2}{R^2}, \tag{2.8}$$

so that the distance scales with the inverse squared transverse momenta of the inputs. The smallest distances, and therefore the first recombinations, are those involving the hardest particles. The clustering is consequently driven by hard inputs. A hard particle accretes the softer inputs around it before soft inputs cluster among themselves.

The name of the algorithm reflects this inversion. In the k_T algorithm, the choice $p = 1$, the distance scales with the positive powers of transverse momentum, so that soft inputs are clustered first and the jet is assembled from the bottom up, following the inverse of the physical showering order. The anti-k_T algorithm reverses this. Taking $p = -1$ turns the soft-first ordering of k_T into a hard-first one. It is, in effect, the k_T algorithm run with the role of soft and hard interchanged, and the prefix “anti” records precisely that inversion. The two sit at opposite ends of the generalised family, with Cambridge/Aachen, insensitive to transverse momentum altogether, between them.

This ordering has a striking geometric consequence. A hard particle with no other hard particle within an angular distance $2R$ collects all softer inputs within a distance R of it, forming a jet that is exactly circular in the (y, ϕ) plane. Where two hard particles lie closer than $2R$, the boundary between their jets is set by their relative transverse momenta, the harder one retaining the larger share, but the jets remain regular in outline. Soft radiation, precisely because it enters the clustering last, cannot deform these boundaries. It is distributed among the hard jets without reshaping them.

A useful way to characterise this stability is through the notion of jet area, the region of the (y, ϕ) plane from which a jet gathers its inputs. This area may be probed operationally by adding a large number of infinitesimally soft “ghost” particles to an event and recording which jet each is assigned to. For most algorithms the resulting areas are irregular and fluctuate from jet to jet, because soft particles influence the clustering. However, for the anti-k_T algorithm, where soft particles are clustered last and cannot reshape the hard jets, the areas are close to the ideal value πR^2 and vary little. A stable, predictable area is what makes the subtraction of pile-up and underlying-event contamination tractable, since that contamination scales with the area over which each jet collects it.

It is this regularity that has made the anti-k_T algorithm the standard jet definition at the LHC. Because the jet boundary is stable against soft radiation, the area over which a jet collects contamination from pile-up and the underlying event is well defined and nearly constant, and the corresponding energy corrections are straightforward to calibrate. The algorithm thus combines the IRC safety guaranteed for the whole generalised-k_T family with a jet shape that is simple to treat experimentally, a combination that its two siblings, whose boundaries are irregular, do not offer.

The prominence of the algorithm is also a matter of timing. When the LHC began taking data, a jet definition was needed that was both IRC safe, so that measurements could be compared with the increasingly precise perturbative calculations then becoming available, and computationally fast enough to run on the very large event samples the machine would produce. The anti-k_T algorithm, together with its efficient implementation in the FastJet package [9], met both needs, and it was adopted as the baseline jet

definition by the LHC experiments, a role it has held since.

For these reasons the anti-k_T algorithm was used in this work to define the jets, with radius parameter $R = 0.8$. Jets constructed in this way are conventionally denoted as AK8. What the algorithm does not provide, however, is a physically meaningful account of how the jet was assembled. Because its sequence is ordered by hardness rather than by angle, the intermediate recombinations bear no correspondence to the sequence of physical splittings in a parton shower, which is ordered in angle. The clustering history of an anti-k_T jet is, in this sense, an artefact of the algorithm rather than a record of the underlying radiation. Recovering that history, which is what the substructure observables of this work require, calls for a different member of the family, applied to the constituents of the jet once it has been defined. That algorithm is the Cambridge/Aachen algorithm and it is the subject of the next subsection.

2.2.3 The Cambridge/Aachen algorithm

By choosing $p = 0$ in equation 2.6 and equation 2.7 the Cambridge/Aachen (C/A) algorithm is defined [10]-[11]. With this value, the transverse momentum factors drop out of the distance measure entirely, which reduces to the purely angular form:

$$d_{ij} = \frac{\Delta R_{ij}^2}{R^2}. \tag{2.9}$$

Clustering with this algorithm proceeds by repeatedly merging the pair of inputs closest in angle, irrespective of how hard or soft they are. Where the anti-k_T algorithm is driven by the hardest inputs, the C/A algorithm orders them by angular seperation alone. This means that the first merges are the most collinear and the last the widest.

This angular ordering is what makes the C/A algorithm the natural tool choice for resolving jet substructure. As set out in Section 2.1, the parton shower is ordered in angle, with the widest angle splittings occuring earliest in the physical branching sequence. A C/A clustering therefore mirrors, in reverse, the order in which the shower developed. Undoing the clustering one step at a time, retraces an approximation to the branching sequence from the widest splitting inwards, and each step exposes a single emision, characterised by the angular seperation ΔR and the relative transverse momentum k_t of the two branches it seperates. It is this correspondence between the C/A history and the physical shower that the Lund jet plane of Section 2.3 is built to exploit, and it is the reason substructure observables are extracted from the C/A history rather than the anti-k_T one.

The declustering procedure can be stated explicitly. Beginning from the single C/A object that contains the whole jet, one undoes the last recombination, splitting it into two subjets a and b, labelled so that $p_{t,a} > p_{t,b}$. The softer branch b is taken as the emission and then follows the harder branch a and repeats, undoing its last recombination in turn. Iterating along the hard branch until it can no longer be declustered traces out the primary sequence of the jet. This is a ordered list of emissions, from the widest and earliest to the most collinear and latest.

Each splitting in this sequence is characterised by a small set of kinematic variables. The angular seperation of the two branches ΔR_{ab}, and their relative transverse momentum $k_t = p_{t,b}\,\Delta R_{ab}$, together locate the emission on the two scales that matter. This is how wide and how hard it is. A third variable, the momentum fraction $z = p_{t,b}/(p_{t,a} + p_{t,b})$, measures how the momentum was shared between emitter and emission. These are the quantities from which the Lund jet plane of section 2.3 is constructed, and

computing them at every declustering step is the immediate goal of the C/A history.

Following only the harder branch at each step captures the primary emissions, but it discards the structure within the softer branches. That structure is recovered by declustering recursively. Following a chosen primary emission, the softer branch b is itself declustered in the same way, exposing the emission within it and generating a secondary sequence of emmissions associated with that primary splitting. The primary sequence records how the leading parton radiated and the secondary sequences record how those radiated partons radiated in turn. Both are needed for the observables of this work, which compare the orientation of a primary splitting with that of the secondary splitting it seeds.

In practice, both the anti-k_{T} and C/A algorithms are applied in sucession. A jet is first defined with the anti-k_{T} algorithm, since it fixes its constituents and its regular boundary. Those constituents are then reclustered with the C/A algorithm, to obtain an angularly ordered history that can expose the emissions within the jet. The radius parameter of the C/A reclustering step was set to $R = 1$, larger than the anti-k_{T} value of $R = 0.8$, to ensure that all the constituents of the original jet are gathered into a single C/A object and that the declustering history is complete with no emmision lost at the edge. The kinematics of the jet are unchanged by this operation, meaning that the same constituents with the same momenta are involved, but they are now organised into a sequence from which the physical splittings can be read.

The division of labour between the two algorithms is thus deliberate. The anti-k_{T} algorithm supplies a jet with a stable, regular boundary, well suited to the energy calibration and pile-up subtraction discussed in Section 2.2.2. The C/A algorithm applied to that jet's constituents supplies a history that mirrors the physical showering sequence with angularly ordered branching history which substructure requires. Neither algorithm alone would suffice. The anti-k_{T} algorithm gives a clean jet but an unphysical history, while the C/A algorithm gives a physical history but an irregular, contamination-prone boundary. Using each where it is strong is what makes the substructure analysis of this work both experimentally robust and physically interpretable.

2.2.4 Flavoured jets and the IFN algorithm

The algorithm of the preceding subsections cluster inputs on the basis of their kinematics alone, and for most purposes that is sufficient. However, for the measurement at the centre of this work, kinematics are not enough. As anticipated in Chapter 1, the goal is to probe the spin correlations of gluon splitting, and that requires distinguishing the channels of Eqs. (2.4), (2.5) and (2.3), which produce, respectively, a pair of gluons, a quark or anti quark with a gluon and a quark-antiquark pair. The channels differ not only in their rates but also their spin interference signal carries opposite signs. Extracting the effect at its true strength therefore requires separating the channels, and that requires flavour assigment to the splittings inside a jet. This raises a question the kinematic clustering algorithms leave unanswered. How can flavour be assigned to a jet, or to a branch within one, in a way that is theoretically well defined and experimentally robust?

The obvious definition is to sum the flavours of a jet's constituents, counting a quark as $+1$ and an antiquark as -1 for each quark flavour seperatelly, so that the net flavour of a jet is that of the parton which initiated it. Simple as it is, this definition is not IRC safe and the source of the problem is precisely the $g \to q\bar{q}$ splitting. A gluon carries no net flavour, but it can split into a quark-antiquark pair that is soft or wide in angle or even both. Were the quark and antiquark then assigned to different jets, or to different branches of the same jet, each would acquire a net flavour that the gluon never possessed. Because the splitting can be arbitrarily soft, the missassigment occurs with a probability that diverges in the same

manner as the emission spectrum of equation (2.2), and no virtual correction is assigned to cancel it. This is the very mechanism identified in Section 2.2.1 for kinematic observables, now reappearing in for flavour. The naive definition is spoiled by exactly soft and collinear radiation that QCD produces most abuntantly, and a fixed order calculation of it fails to converge. Flavour assigned carelessly is therefore not a physical quantity, and it cannot be compared with theory.

Restoring IRC safety to jet flavour has been pursued for two decades. The earliest safe prescriptions achieved it by making the clustering distance itself depend on the flavour of the objects being combined [12], which succeeded in taming the divergence but at a price of the resulting jets not having the kinematics of the underlying algorithm. This means that a flavour aware jet and a standard anti-k_T or C/A jet were no longer the same object. For a measurement that depends on both the kinematics of a jet and flavour of its structure, that price is too high. The approach adopted in this work, Interleaved Flavour Neutralisation (IFN) [13], was designed precisely to avoid it. IFN leaves the kinematics of the clustering entirely untouched and treats flavour as a seperate layer that runs alongside the ordinary sequence. The jets returned by this algorithm are exactly identical to the C/A jets of Section 2.2.3, constituent for constituent and momentum for momentum. What IFN supplies in addition is a flavour label for each, computed by a procedure that is stable against soft and collinear emission.

The principle behind that procedure is neutralisation. As the clustering proceeds, IFN follows the flavour carried by every object in the sequence, and, at each stage, asks wether a flavoured object can be matched to a nearby object of opposite flavour, the partner with which it would have originated in a common $g \to q\bar{q}$ splitting. When such a pair is found, the two flavours are neutralised against each other, so that the spurious net flavour is cancelled at the moment it would otherwise be passed up the clustering tree and inherited by a jet. A quark and antiquark born of the same soft or collinear gluon splitting are in this way reunited in flavour, even though their kinematics remain wherever the C/A algorithm placed them, and the divergence that defeated the naive definition is removed order by order. The matching is governed by a flavour distance that weighs how soft and how collinear a candidate pair is, so that it is exactly the dangerous soft and collinear configurations that are neutralised, while genuinely distinct hard flavours are left in place.

What distinguishes IFN from a prescription that merely labels the final jet is that the flavour information is exposed at every step of the clustering and not only at the end. This is the feature that makes the present analysis possible. The clustering of a C/A history produces at each step a splitting of one object into two branches. IFN attaches to those branches a flavour label that is IRC safe and, therefore, meaningful at fixed order in perturbation theory. The abstract distinction between the discussed channels in Eqs. (2.4), (2.5) and (2.3) is thereby made concrete, and it can be read off at each splitting in the history of a jet. It is on this per splitting flavour that the channel selection at the heart of the measurement is built. The algorithm is distributed as a plugin to the FastJet package [9], and the configuration adopted here, together with the way its per-splitting labels are combined into the selection itself, is set out in Chapter 3.

2.3 The Lund jet plane

The declustering of the Cambridge/Aachen history, described in Section 2.2.3, associates with every jet an ordered sequence of emissions, each carrying an angular seperation ΔR, a relative transverse momentum k_t and a momentum fraction z. On its own, this sequence is an unwieldly list of numbers. Jets have different numbers of emissions and the raw variables span many orders of magnitude. The Lund jet plane [1] is the representation that turns this list into something legible, by mapping each emission to

a single point in a two-dimensional plane whose coordinates are chosen so that the structure of QCD radiation is exposed in its simplest form.

2.3.1 Coordinates and the uniform density

The construction of the Lund jet plane is direct. For each jet, the primary declustering of the C/A history, of Section 2.2.3, yields a list of emissions. Every emission in that list is mapped to a single point on the plane, using its angular separation ΔR and relative transverse momentum k_t as coordinates. Repeating this for every emission in the sample and overlaying the resulting points produces a two-dimensional distribution, which is then normalised to the number of jets to give the average density of emissions per unit area. It is this averaged distribution, rather than the pattern of any single jet, that is referred to as the Lund jet plane and that carries the statistical information exploited below.

Each primary emission is placed on the plane at the coordinates:

$$\left(\ln \frac{1}{\Delta R},\, \ln k_{\mathsf{T}}\right), \tag{2.10}$$

the horizontal axis measures the inverse angle of the splitting and the vertical axis its hardness, both logarithmically. The choice of logarithmic axes is not cosmetic. Recalling in Section **??**sec:qcd) that the emission probability of a gluon emission carries the double structure $d\mathcal{P} \propto \alpha_s\, d\theta/\theta\; dE/E$, which is one factor for the collinear enhancement and one for the soft enhancement. Written in terms of the Lund coordinates of equation (2.10), each of these factors becomes flat. The measure $d\theta/\theta$ is simply $d\ln(1/\Delta R)$, and the energy factor maps onto $d\ln k_{\mathsf{T}}$. The doubly divergent emission spectrum of the raw variables therefore becomes, on the Lund plane, an approximately uniform density of emissions per unit area,

$$\rho \;=\; \frac{1}{N_{\text{jet}}}\,\frac{dn_{\text{emissions}}}{d\ln(1/\Delta R)\; d\ln k_{\mathsf{T}}} \;\approx\; \text{const} \times \frac{\alpha_s}{\pi}, \tag{2.11}$$

up to the slow running of α_s and the effects discussed below. It is this flatness that makes the plane so useful against a uniform background. Any physical effect that departs from the leading soft-collinear picture stands out as a local excess or depletion, rather than being buried in the steeply falling raw spectrum. This uniform density is precisely what is observed in practice. Figure 2.1 shows the average primary Lund plane obtained from a simulated sample of QCD jets. Across the perturbative bulk of the plane the density is approximately constant, as anticipated by equation (2.11), falling away only towards the boundaries of the kinematically accessible region.

2.3.2 Regions of the plane

The accessible area of the plane is bounded, and its interior divides into regions of distinct physical origin, sketched in Figure 2.2. The upper boundary is set by the kinematics of the jet. An emission cannot be harder than the jet which contains it, so it caps $\ln k_{\mathsf{T}}$ from above. Along the right of the plane, at large $\ln(1/\Delta R)$, lie the collinear emissions, those radiated at small angle to their emitter. Towards the bottom, at small $\ln k_{\mathsf{T}}$, lie the soft emissions. The perturbative description of Section 2.1 holds across the bulk of this region, where k_{T} is large enough for α_s to remain small; the density there is the near-constant value

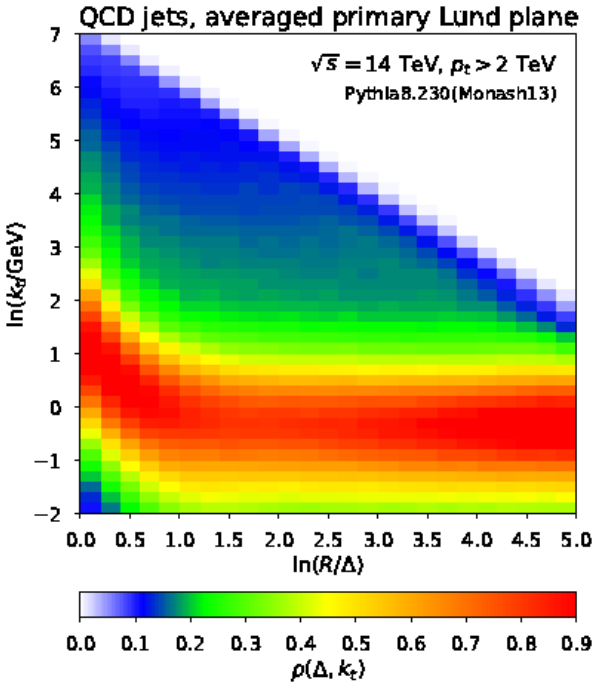


Figura 2.1 – The average primary Lund plane density, ρ, for jets clustered with the C/A algorithm and $R = 1$ having $p_t > 2$ TeV and $|y| < 2.5$, in a simulated QCD dijet sample [1].

of equation (2.11).

The exception is the lower-right corner, at both small angle and low transverse momentum. There k_T falls to the scale of a few hundred MeV, α_s grows large, and the perturbative description breaks down. This is the non-perturbative region, populated by hadronisation and by the soft, wide-angle contamination of the underlying event. One of the strengths of the Lund representation is that these non-perturbative effects are confined to a localised corner of the plane, cleanly separated from the perturbative bulk, rather than being smeared across the whole observable. A measurement can therefore be restricted, by eye or by an explicit cut, to the region where theory applies.

2.3.3 Grooming and the primary and secondary planes

This seperation motivates grooming. Grooming is the deliberate removal of soft, wide-angle emissions that dominate the non-peturbative region of the plane. The grooming procedure used in this work is Soft Srop [14], which walks through the declustering and, at each step, retains the harder branch unless the softer one satisfies:

$$z > z_{\text{cut}} \left(\frac{\Delta R}{R} \right)^{\beta}, \tag{2.12}$$

in which case both branches are kept and the declustering continues. The parameter z_{cut} sets how soft an emission must be to be discarded, and β controls the angular reach of the grooming. The choice $\beta = 0$, adopted here for the Soft Drop, makes the condition purely one of momentum fraction, $z > z_{\text{cut}}$, independent of angle. In this limit, Soft Drop reduces to the modified Mass-Drop Tagger [15]. On the Lund plane the effect of grooming is immediate and visual. It removes the emissions in the soft, low k_T band, clearing the non-peturbative corner and leaving a plane populated only by the harder splittings.

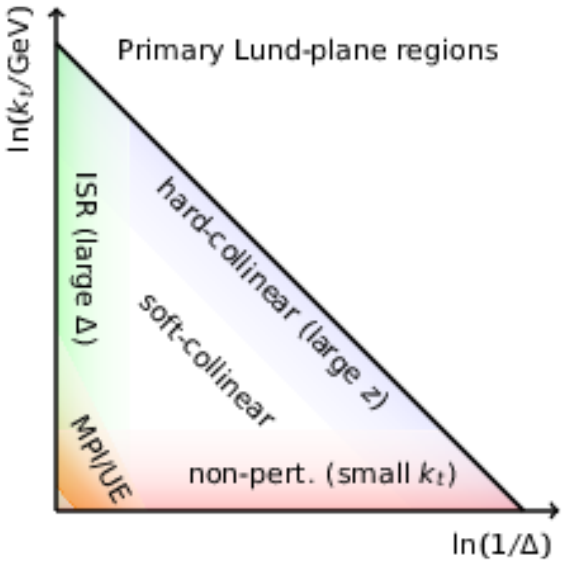


Figura 2.2 – Schematic representation of the di□erent regions of the Lund plane [1].

In this work, two versions of the plane are produced and used as follows. The primary Lund plane collects the emissions along the hard branch of the jet, one point per jet splitting, and describes how the leading parton radiated. The secondary Lund plane is built by taken the chosen primary emission and declustering its softer branch in turn, as described in Section 2.2.3. Its points describe how that radiated parton itself went on to radiate. The two planes carry complementary information, and both are needed for the spin observables of Section 2.4, which compare the orientation of a primary splitting with that of the secondary splitting it seeds. Figure 2.3 illustrates these two levels of structure for two example jets. The middle row shows the full Lund diagram, in which every emission, primary and secondary, is represented, while the bottom row shows the primary Lund plane alone, retaining a single point for each primary splitting.

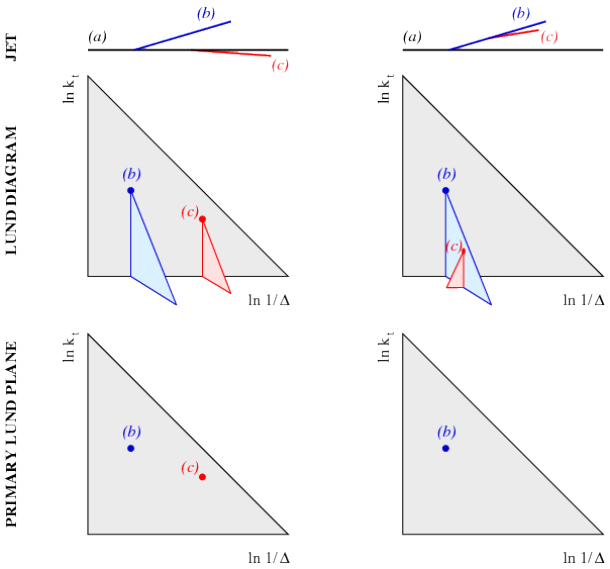


Figura 2.3 – Different representations for two jets. Top: the particles inside the jet. Middle: the full Lund diagram. Bottom: the primary Lund plane [1].

2.4 Collinear spin correlations

The tools assembled in the preceding sections have so far been described as though a parton shower were a sequence of independent splittings, each each blind to those before it. That description is incomplete. The intermediate partons in a shower carry spin, and their spin in correlated with the kinematics of the splitting that produces them. When such a parton subsequently branches, that correlation is transmitted to the azimuthal orientation of its own decay products. Successive splittings are therefore not independent, meaning that the plane of one branching is correlated with the plane of the next. These are the collinear spin correlations that this work sets out to measure, and they are the subject of this section.

This effect is a genuine quantum interference phenomenon, with no counterpart in a classical picture of sequential radiation, and it manifests in a distinctive and robust way. The outcome appears as a modulation of the azimuthal angle between successive splitting planes with a period of π rather than 2π, that is, as a dependence on $\cos 2\psi$. This section develops that statement in four steps. Section 2.4.1 sets out the physical mechanism and the origin of the $\cos 2\psi$ modulation. Section 2.4.2 shows how the size and sign of the modulation depend on the splitting channel, the property that makes flavour identification essential to the measurement. Sections 2.4.3 and 2.4.4 then introduce the two observables through which this work accesses the effect. These observables are the Lund declustering angle $\Delta\psi_{12}$, built on the branching history of Section 2.3, and the three-point energy correlator, which reaches the same physics through the flow of energy within the jet.

2.4.1 Mechanism and the $\cos 2\psi$ modulation

The origin of the correlation lies in the spin of the intermediate gluon [16]. Consider a gluon produced in a first splitting and undergoing a second splitting of its own, the configuration that the primary and secondary Lund planes of Section 2.3 isolate. The gluon is a spin-one particle, but a hard, nearly on-shell gluon has only two physical polarisation states, transverse to its direction of motion. In the collinear limit, these two states are most naturally described not as circular polarisations but as two orthogonal linear polarisations, each an oscillation of the field along a fixed axis in the plane transverse to the gluon momentum.

The polarisation of the intermediate gluon is not random. The first splitting, which produced it, selects a preferred transverse axis. The gluon emerges linearly polarised, with its polarisation correlated with the plane of that first branching. This is the essential point. A gluon radiated in a splitting carries a memory of the orientation of that splitting, encoded in the direction of its linear polarisation.

That memory is then read out by the second splitting. The probability that a linearly polarised gluon branches into a given final state depends on the azimuthal orientation of the second splitting plane relative to the polarisation of the axis. Writting ψ for the angle between the first and the second splitting planes, the differential rate acquires a dependence on ψ through this alignment, and the two splittings become correlated in their relative azimuth.

The period of that correlation is fixed by the nature of the polarisation. A linear polarisation is specified by an axis, not by a direction along it. Rotating the polarisation by π returns the same physical state, since an oscillation along an axis is unchanged by reversing its sense. The azimuthal dependence must therefore be invariant under $\psi \to \psi + \pi$, which forbids a $\cos\psi$ term and admits $\cos 2\psi$ as its leading harmonic. This is the distinctive signature of the effect. The result has to have a modulation of period π rather than 2π, reflecting the headless, axial character of linear polarisation. Collinear spin correlations

announce themselves not as any azimuthal dependence, but specifically as a $\cos 2\psi$ dependence.

At second order in the strong coupling and in the collinear limit, the cross section differential in ψ therefore takes the form

$$\frac{d\sigma}{d\psi} \propto a_0 \left(1 + \frac{a_2}{a_0} \cos 2\psi\right), \tag{2.13}$$

where a_0 is the spin-averaged rate, or, in other words, the result one would obtain by ignoring the polarisation of the intermediate gluon, and a_2 is the coefficient of the interference term [16]. The ratio a_2/a_0 measures the strength of the correlation where it vanishes if the intermediate gluon carries no polarisation memory, and its magnitude sets the depth of the $\cos 2\psi$ modulation. Its sign determines the phase, wether the second splitting preferentially aligns with the first splitting plane, giving maxima at $\psi = 0, \pi$, or preferentially lies perpendicular to it, giving maxima at $\psi = \pm\pi/2$. The measurement of these spin correlations reduces, in essence, to the measurement of this single ratio, and the whole of the analysis that follows is directed towards extracting it cleanly.

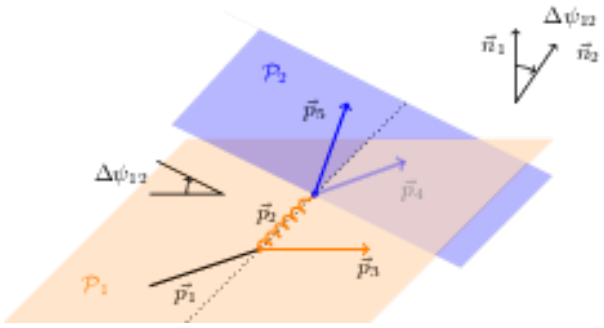


Figura 2.4 – Azimuthal angles are defined between successive splitting planes for the $1 \to 23$ splitting, $\mathcal{P}_1 \supset \{\vec{p}_2, \vec{p}_3\}$, and the $2 \to 45$ splitting, $\mathcal{P}_2 \supset \{\vec{p}_4, \vec{p}_5\}$. The figure also depicts the vectors normal to the two planes, $\vec{n}_1$ and $\vec{n}_2$ [16].

2.4.2 Channel dependence of a_2/a_0

The mechanism of Section 2.4.1 was described for a generic intermediate gluon, but the strength and sign of the resulting modulation are not universal. They depend on the nature of the second splitting, or, that is, on the channel through which the polarised gluon decays. The two channels available to it are those of Eqs. (2.4) and (2.5). The gluon may split into a pair of gluons, $g \to gg$, or into a quark–antiquark pair, $g \to q\bar{q}$. Each reads out the polarisation of the parent gluon, but the two do so with opposite sign [16].

The origin of the difference is the spin of the produced particles. The angular distribution of a splitting depends on how the daughters' spins couple to the polarisation axis of the parent, and this coupling differs between a final state of two vector gluons and one of two spin-half fermions [16]. The consequence is that the interference coefficient a_2/a_0 of equation (2.13) enters with one sign for $g \to gg$ and the opposite sign for $g \to q\bar{q}$, where one channel favours alignment of the two splitting planes, the other favours a perpendicular configuration.

This sign difference is the central obstacle that the present work is built to overcome. A measurement that does not distinguish the channel of the second splitting sums over both, and the two contributions, carrying opposite signs, partially cancel. What survives in such an inclusive measurement is not the intrinsic strength of the correlation but a residue, its magnitude and even its sign set by the accidental balance of $g \to gg$ and $g \to q\bar{q}$ splittings in the sample.The interference is not absent. It is diluted, and the

degree of dilution reflects the flavour composition of the selection rather than any property of QCD.

It follows that a faithful measurement of the effect requires isolating a single channel. Selecting the $g \to q\bar{q}$ splittings, and measuring a_2/a_0 on that subsample alone, replaces the diluted inclusive residue with the genuine coefficient of a single channel. This is precisely what the infrared-safe and collinear-safe flavour labelling of Section 2.2.4 makes possible. By tagging the flavour of each splitting exposed in the declustering, it allows the $g \to q\bar{q}$ channel to be enriched and the $\cos 2\psi$ modulation to be measured as a function of the purity of that enrichment. The channel dependence described here is therefore not merely a feature of the physics but the reason the flavour selection developed in this work is indispensable, and it is the thread that connects the spin observable of the next section to the flavoured-jet algorithm of Section 2.2.4.

2.4.3 The Lund observable $\Delta\psi_{12}$

The mechanism of the preceding sections can be turned into a concrete observable by using the Lund declustering of Section 2.3 to identify the two splittings whose planes are to be compared. This is the construction proposed in Ref. [16], and the observable it defines is the azimuthal angle $\Delta\psi_{12}$ between the hardest primary and secondary branchings of a jet.

The two splittings are selected as follows. Declustering the primary Lund sequence of the jet yields a set of primary emissions, from which, among those who pass the Soft Drop condition of equation (2.12), the one with largest relative transverse momentum k_T is taken as the first splitting. Its softer branch is then declustered in turn, generating the secondary sequence associated with it. Equally to the procedure of choosing the frist splitting, among the secondary emissions that pass the same condition, the hardest in k_T is taken as the second splitting.Each of the two chosen splittings defines a plane in momentum space. The plane $\mathcal{P}_1$ spanned by the branches of the first, and $\mathcal{P}_2$ spanned by the branches of the second. With this, $\Delta\psi_{12}$ is the azimuthal angle between them, illustrated in Figure 2.4.

By construction, $\Delta\psi_{12}$ is the realisation for jets of the abstract angle ψ of equation (2.13). Its distribution therefore inherits the $\cos 2\psi$ modulation derived in Section 2.4.1,

$$\frac{1}{\sigma}\frac{d\sigma}{d\Delta\psi_{12}} \;\propto\; 1 + \frac{a_2}{a_0}\cos 2\Delta\psi_{12}, \tag{2.14}$$

and the coefficient a_2/a_0 extracted from it is the quantity that carries the spin-correlation physics. Two features of the definition are essential to what follows. First, because the second splitting is drawn from the secondary sequence seeded by the first, $\Delta\psi_{12}$ compares precisely the two successive branchings between which the polarisation memory of Section 2.4.1 is transmitted. Second, the second splitting is a gluon decay whose channel $g \to gg$ or $g \to q\bar{q}$ can be identified by the flavour labelling of Section 2.2.4. It is this second feature that makes $\Delta\psi_{12}$ not merely a measurement of the inclusive, diluted modulation, but a channel-resolved probe of the effect. Through restricting to secondary splittings tagged as $g \to q\bar{q}$, the sign cancellation of Section 2.4.2 is undone, and a_2/a_0 can be measured on a single channel.

2.4.4 The three-point energy correlator

The Lund observable of the previous section is not the only route to the collinear spin correlations of Section 2.4.1. The same $\cos 2\psi$ modulation can be accessed through the energy flow within a jet, without

reference to any clustering history, using the three-point energy correlator (EEEC) [17]. Where $\Delta\psi_{12}$ reconstructs the two splitting planes by declustering, the energy correlator reads the effect directly from the angular and energy distribution of the jet's constituents, and the two therefore provide complementary handles on the same underlying physics.

The three-point energy correlator measures the energy-weighted distribution of triplets of particles as a function of their relative angles. Concretely, it is defined as the energy-weighted sum over all triplets of particles in the jet,

$$\frac{1}{\sigma}\frac{d^3\Sigma}{d\psi\, d\theta_S\, d\theta_L} = \sum_{i,j,k} \frac{p_{\mathrm{T},i}\, p_{\mathrm{T},j}\, p_{\mathrm{T},k}}{p_{\mathrm{T,jet}}^3}\, \delta(\psi - \psi_{ijk})\, \delta(\theta_S - \theta_{S,ijk})\, \delta(\theta_L - \theta_{L,ijk}), \tag{2.15}$$

where the sum runs over all triplets (i, j, k) of the jet's constituents, each weighted by the product of their transverse momenta normalised to that of the jet. Here θ_S and θ_L are the smallest and largest opening angles of the triplet, and ψ is the azimuthal angle between the plane spanned by the two closest particles and that spanned by the third. Its power for the present purpose is that this last variable retains sensitivity to the azimuthal orientation discussed in Section 2.4.1. This means that the correlator's dependence on ψ carries the same $\cos 2\psi$ modulation, governed by the same interference between gluon polarisation states. Being built from energy weights and angles alone, the correlator is infrared and collinear safe by construction and requires no declustering, no grooming and no jet-flavour assignment.

This makes the energy correlator an independent cross-check on the physics extracted through $\Delta\psi_{12}$. It reaches the same $\cos 2\psi$ effect by a route that shares none of the reconstruction choices on which the Lund observable depends, so that agreement between the two would establish that the measured modulation is a property of the radiation rather than an artefact of any one analysis chain. Note that the coefficient a_2/a_0 extracted from the energy correlator is not expected to coincide numerically with that from $\Delta\psi_{12}$, since the two observables define the azimuthal angle through different objects and carry different sensitivities to the kinematic selection. What the two share is the sign and the qualitative behaviour dictated by the underlying gluon-polarisation interference, and it is at that level that they cross-check one another. The correlator is developed in this work as a complementary probe alongside the primary $\Delta\psi_{12}$ measurement, and its implementation and results are presented in Chapter 4.

3 METHODS

4 RESULTS AND DISCUSSION

5 CONCLUSIONS

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